SMART KISAN

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

Vineet Patidar EN23CS3011132

Devansh Mishra EN24CS3T10010

Under the Guidance of

Prof. Kumar Gaurav

Prof. Shilpa Raghuvanshi

Department of Computer Science & Engineering

Faculty of Engineering

MEDICAPS UNIVERSITY, INDORE- 453331

APRIL-2026

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Report Approval

The project work **"Smart Kisan"** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the "Project Report" only for the purpose for which it has been submitted.

Internal Examiner

Name:

Designation

Affiliation

External Examiner Name:

Designation

Affiliation

Declaration

I/We hereby declare that the project entitled **"Smart Kisan"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications in 'Computer Science and Engineering' completed under the supervision of **Prof. Kumar Gaurav** and **Prof. Shilpa Raghuvanshi,** Faculty of Engineering, Medi-Caps University Indore is an authentic work.

Further, I/we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

Certificate

I/We, **Prof. Kumar Gaurav** and **Prof. Shilpa Raghuvanshi** certify that the project entitled **"Smart Kisan"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications by **Vineet Patidar** and **Devansh Mishra** is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

_____

Prof. Kumar Gaurav and **Prof. Shilpa Raghuvanshi**

Faculty of Engineering

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Dr. Ratnesh Litoriya

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Acknowledgements

I would like to express my deepest gratitude to the Honorable Chancellor, **Shri R C Mittal,** who has provided me with every facility to successfully carry out this project, and my profound indebtedness to **Prof. (Dr.) D. K. Patnaik,** Vice Chancellor, Medi-Caps University, whose unfailing support and enthusiasm has always boosted up my morale. I also thank **Dr. Ratnesh Litoriya,** Dean, Faculty of Engineering, Medi-Caps University, for giving me a chance to work on this project. I would also like to thank my Head of the Department **Prof. Kailash Chandra Bandhu** for his continuous encouragement for the betterment of the project.

It is their help and support, due to which we became able to complete the design and technical report.

Without their support this report would not have been possible.

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Abstract

Agriculture remains the backbone of India's economy, employing more than 50% of the workforce and contributing significantly to the nation's GDP. However, Indian farmers continue to face formidable challenges including unpredictable weather, soil degradation, limited access to agronomic expertise, crop disease outbreaks, and volatile market prices. The absence of timely, datadriven guidance leads to poor crop selection, suboptimal resource utilization, and avoidable postharvest losses.

SMART KISAN is an AI-powered, multi-feature smart agriculture web application developed using Python and Streamlit to bridge this information gap. The platform consolidates five core agricultural intelligence modules into a single, user-friendly interface: (1) AI-driven Crop Recommendation that leverages soil parameters (N, P, K, pH), climatic variables (temperature, humidity, rainfall) and location data to suggest the most suitable crop; (2) Crop Advisory, a comprehensive knowledge base covering 35+ crops across all major Indian agricultural categories including soil requirements, cultivation schedules, fertilizer management, pest control, and postharvest practices; (3) Yield Estimator that predicts crop output based on regional historical data and current agronomic conditions using a trained machine learning model; (4) Profit Calculator to help farmers make informed economic decisions by projecting net returns against input costs; and (5) Market Price Intelligence that provides real-time and historical Minimum Support Price (MSP) and mandi rates across Indian states.

The system integrates an EfficientNet-B0 convolutional neural network (pre-trained on ImageNet) as the backbone for the crop recommendation and disease detection pipeline, offering high accuracy with low computational overhead. The crop advisory database covers 35 crops spanning cereals, pulses, oilseeds, fiber crops, spices, fruits, and vegetables, with each crop entry containing over 25 structured data attributes.

SMART KISAN was built with scalability and accessibility in mind, deployable via Streamlit Community Cloud without requiring specialized infrastructure. Evaluation results demonstrate high user satisfaction, accurate crop recommendations, and meaningful yield estimation. This project represents a practical step toward precision agriculture in India, contributing to the United Nations Sustainable Development Goal 2 (Zero Hunger) and SDG 8 (Decent Work and Economic Growth).

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
API	Application Programming Interface
MSP	Minimum Support Price
NPK	Nitrogen, Phosphorus, Potassium
IoT	Internet of Things
UI	User Interface
N, P, K	Nitrogen, Phosphorus, Potassium (soil nutrients)
FYM	Farm Yard Manure
SDG	Sustainable Development Goal
GDP	Gross Domestic Product
ICAR	Indian Council of Agricultural Research
KNN	K-Nearest Neighbors
RF	Random Forest
SVM	Support Vector Machine
Kharif	Monsoon season crop (June-November)
Rabi	Winter season crop (November-April)
Zaid	Summer season crop (March-June)

Chapter 1: Introduction

1.1 Introduction

India is predominantly an agrarian economy. With more than 1.4 billion people and over 600 million engaged directly or indirectly in agriculture, the sector contributes approximately 18% to the national GDP and remains crucial for food security. Yet, Indian agriculture is beset by a multitude of structural challenges: fragmented landholdings, over-dependence on monsoon rainfall, erratic weather patterns driven by climate change, poor soil health, inadequate cold-chain infrastructure, and a widening information asymmetry between urban agricultural scientists and rural farmers.

A farmer in rural Madhya Pradesh or Maharashtra, with limited formal education, must make high-stakes decisions every season: Which crop to sow? How much fertilizer to apply? When to irrigate? What are the prevailing market prices? These decisions, if made poorly, can result in catastrophic crop failure, severe debt, and lasting socio-economic distress. In extreme cases, financial ruin has tragically culminated in farmer suicides, a crisis that has claimed more than 300,000 lives in India over the past two decades.

The proliferation of affordable smartphones, budget mobile data plans, and digital literacy initiatives has created an unprecedented opportunity. Over 400 million rural Indians now own smartphones, and agri-tech adoption is growing rapidly. Artificial Intelligence and Machine Learning have matured to the point where they can be deployed effectively even on constrained hardware. The convergence of these trends forms the foundation of SMART KISAN.

SMART KISAN — the name translates to 'Smart Farmer' in Hindi — is designed as a comprehensive, AI-powered digital farming companion. It does not replace the farmer's judgment; rather, it equips the farmer with the data-driven insights and agronomic knowledge that were previously accessible only to large agribusinesses or well-connected individuals. By democratizing agricultural intelligence, SMART KISAN aims to meaningfully improve the livelihoods of millions of farming households across India.

1.2 Literature Review

Digital agriculture systems have evolved from simple database lookup tools into more intelligent decision-support platforms. Early systems mainly stored crop data and displayed static

cultivation advice. Later systems introduced rule-based engines that mapped soil type or climate zone to suitable crops. More recent tools use machine learning, remote sensing, and image analysis to predict crop suitability, estimate yield, and identify diseases.

Crop recommendation research typically uses features such as rainfall, temperature, humidity, soil profile, irrigation type, and season. A model is trained on historical farm records or curated agricultural datasets to learn which crops perform well under which environmental conditions. Yield prediction research adds another layer by estimating output after a crop has been selected.

Another important branch of research is market intelligence. Even if two crops are agronomically suitable, the final choice depends on market demand, price stability, transport cost, and seasonal volatility. For this reason, modern farm advisory tools increasingly integrate price feeds or local reference prices.

1.3 Objectives

The primary objectives of the SMART KISAN project are:

1. To develop an AI-powered crop recommendation engine using machine learning models trained on soil and climatic parameters.
2. To build a structured, searchable crop advisory knowledge base covering 35+ crops with granular agronomic data.
3. To implement a yield estimation module using regression-based machine learning trained on regional agricultural statistics.
4. To create an economic profit calculator enabling farmers to make informed financial decisions pre-sowing.
5. To provide a market price intelligence module displaying current and historical crop prices by state.
6. To deploy the application as an accessible, low-cost Streamlit web application suitable for smartphone use.

1.4 Significance

The significance of **SMART KISAN** lies in its ability to bridge the gap between technical prediction models and the everyday, practical questions faced by farmers. Traditional advisory systems often provide recommendations without explaining the reasoning behind them, which can leave farmers uncertain about whether to trust or adopt the advice. SMART KISAN addresses this challenge by not only offering recommendations but also presenting the underlying rationale. By integrating crop profiles, soil data, irrigation classifications, and market references, the application

delivers a holistic perspective that empowers farmers to make informed decisions rather than relying on isolated outputs.

Another important aspect of this project is its accessibility. SMART KISAN leverages widely available software tools and open-source Python libraries, making it reproducible in academic environments such as college laboratories and adaptable for small-scale pilot deployments. This ensures that the system does not require expensive infrastructure, thereby lowering barriers to adoption and enabling experimentation in resource-constrained settings.

Beyond its technical merits, the project holds social and economic significance. By equipping farmers with transparent, data-driven insights, SMART KISAN fosters trust in technology and encourages sustainable farming practices. It can help reduce crop failures, optimize resource usage, and improve profitability by aligning agricultural decisions with both environmental conditions and market trends. Furthermore, its modular architecture allows for future scalability, meaning additional datasets or predictive models can be incorporated as the system evolves.

In essence, SMART KISAN is not just a technological tool but a practical companion for farmers. Its significance lies in democratizing advanced agricultural analytics, making them understandable, affordable, and actionable for communities that need them most.

1.5 Research Design

The research design of this project is **applied and implementation-oriented**, focusing on practical usability rather than abstract theory. Instead of developing a model in isolation, the project begins with **user needs**—the everyday questions and challenges faced by farmers—and systematically maps those needs to corresponding software modules. This ensures that the design remains grounded in real-world agricultural practices while leveraging modern computational tools.

The design follows a **layered architecture** to maintain clarity, modularity, and scalability:

- **Presentation Layer**: Built with Streamlit, this layer manages the user interface, ensuring that farmers and researchers can interact with the system through intuitive pages and dashboards.

- **Application Layer**: This layer contains the prediction logic, algorithms, and calculations. It translates raw data into actionable insights, such as crop recommendations or irrigation classifications.
- **Data Layer**: Responsible for storing structured information, including crop profiles, soil dictionaries, and market references. It acts as the knowledge base of the system.

From a **software perspective**, the design emphasizes **code reuse and modularity**. Utility modules such as the soil estimator, water classifier, and shared constants are designed to be reusable across different parts of the application. This reduces redundancy, improves maintainability, and makes the system easier to extend in future iterations.

1.6 Source of Data

The project draws upon a diverse set of data sources to ensure that SMART KISAN provides accurate, context-aware, and actionable insights. The primary repositories include the local crop database, crop-specific modules stored in data/crops, state conversion values, and trained model files located under ml/models. These sources form the backbone of the system, enabling it to deliver predictions and recommendations tailored to specific agricultural scenarios. The market-price component is designed with flexibility in mind. It can operate using locally stored reference values for offline use, while also supporting API-based retrieval when internet connectivity is available. This dual approach ensures that farmers can access market insights regardless of infrastructure limitations. Similarly, the crop advisory page relies on image assets stored in static/images/crops, while disease-related content references images in static/images/diseases. These visual resources enhance usability by making the advisory system more intuitive and farmer-friendly.

A key strength of the project lies in its intentional use of multiple data formats.

Dictionaries are employed for simple lookups, such as soil type conversions or crop-to-state mappings, ensuring fast and lightweight retrieval.

Serialized encoders and scalers are used for repeated model predictions, allowing machine learning components to operate efficiently without retraining.

- **Modules and artifacts** provide crop-specific intelligence, combining rule-based knowledge with predictive analytics.

This hybrid design makes SMART KISAN a blend of **rule-based information retrieval** and **data-driven prediction**. Rule-based components guarantee consistency and transparency, while predictive models introduce adaptability and learning from historical patterns. Together, they create a system that is both reliable and dynamic.

1.7 Chapter Scheme

Chapter 1 introduces the problem, objectives, significance, and data sources.

Chapter 2 explains the user profile, requirements, dependencies, and assumptions.

Chapter 3 presents the system design, algorithms, diagrams, and database structure.

Chapter 4 discusses implementation technologies, testing, installation, and end-user

Chapter 5 describes the visible results, user interface, modules, and backend data

Chapter 6 summarizes the work and the main conclusions drawn from the project

Chapter 7 outlines future enhancements and possible expansion directions.

Chapter 2: Requirements Specification

The requirements of SMART KISAN are shaped by the kind of user the project is intended for. The main user is a farmer or agriculture learner who wants quick decision support without having to understand the internal logic of machine learning. A secondary user may be a student, trainer, or project evaluator who wants to inspect how the system organizes data and how each module contributes to the final output.

The interface therefore uses clear labels, dropdowns, number inputs, buttons, and result cards. The user is expected to complete an action in a few steps: choose inputs, press a button, and read the result. This reduces training effort and makes the prototype friendly for demonstrations.

2.1 User Characteristics

Requirements Specification The expected users of SMART KISAN fall into two broad categories, each with distinct needs and interaction styles.

1. Farmers and Agricultural Practitioners (Non-technical or Semi-technical Users) This group represents the primary audience of the application. They may possess practical knowledge about crop names, common soil types, rainfall patterns, and local market behavior, but they are unlikely to have expertise in handling datasets, coding, or machine-learning pipelines. For them, the interface is designed to be **simple, intuitive, and highly visual**.

- **Interface Features**: Clear labels, dropdown menus, number inputs, buttons, and result cards ensure that users can navigate the system without technical training.
- **User Needs**: Quick access to crop recommendations, irrigation guidance, and market references presented in a way that is easy to understand.
- **Design Consideration**: The system avoids jargon and emphasizes transparency, showing not only the recommendation but also the reasoning behind it. This builds trust and encourages adoption among farmers.

2. Evaluators, Researchers, and Developers (Technical Users) A secondary group of users includes evaluators, agricultural researchers, and software developers. Their focus is less on day-to-day farming decisions and more on the **system's architecture, modularity, and scalability**.

- **Interface Features**: Access to module separation, code organization, and backend logic for verification and improvement.
- **User Needs**: Ability to test predictive accuracy, validate data sources, and assess whether the codebase can support future enhancements.
- **Design Consideration**: Documentation, modular coding practices, and reusable utility functions (such as soil estimators and water classifiers) are emphasized to make the system extensible and adaptable.

Hybrid Design for Diverse Users By accommodating both non-technical farmers and technical evaluators, SMART KISAN ensures inclusivity. Farmers benefit from a user-friendly interface that

simplifies complex predictions, while developers and researchers gain a structured environment for experimentation and improvement. This dual focus strengthens the project's relevance across practical and academic domains.

2.2 Functional Requirements

The table below lists all functional requirements identified for Smart Kisan:

Req. ID	Description	Module	Priority
REQ-1	**Accept user inputs: state, soil type, sowing month, water availability, irrigation method.**	**Crop Recommendation**	**High**
REQ-2	**Auto-estimate NPK values from state-soil-season combination.**	**Soil NPK Estimator**	**High**
REQ-3	**Classify water availability into 5 categories (Very Low – Very High).**	**Water Classifier**	**High**
REQ-4	**Run Random Forest Classifier and return top-3 recommended crops with confidence scores.**	**Crop Recommender**	**High**
REQ-5	**Validate all user inputs before prediction.**	**All Modules**	**High**
REQ-6	**Accept land size and year for yield estimation.**	**Yield Estimation**	**High**
REQ-7	**Run Gradient Boosting Regressor and return predicted yield in quintals.**	**Yield Estimation**	**High**
REQ-8	**Fetch real-time market prices via Data.gov.in API with 5-min TTL cache.**	**Market Price Fetcher**	**High**
REQ-9	**Calculate expected profit: yield × price – input/labour/irrigation costs.**	**Profit Calculator**	**High**
REQ-10	**Display comprehensive crop advisory: fertilizer, irrigation plan, pest management, harvest.**	**Crop Advisory**	**Medium**
REQ-11	**Identify diseases/pests from uploaded images using EfficientNet-B0.**	**Disease Prediction**	**Medium**
REQ-12	**Display a navigation dashboard with all modules accessible.**	**Landing Page**	**Medium**
REQ-13	**Store and restore session state across page navigations.**	**Session State**	**Medium**

2.3 Dependencies

Requirements Specification The SMART KISAN project relies on a combination of **software libraries, runtime assets, and external resources** to deliver its functionality. These dependencies are carefully chosen to balance usability, performance, and reproducibility.

Software Dependencies

- **Streamlit**: Provides the user interface framework, enabling interactive pages, dropdowns, buttons, and result cards.
- **pandas and NumPy**: Handle structured calculations, data manipulation, and numerical operations essential for agronomic inputs and profit calculations.
- **scikit-learn and joblib**: Support machine learning model loading, preprocessing, and serialization, ensuring that trained models can be reused without retraining.
- **XGBoost**: Enhances predictive performance by offering gradient boosting models for yield estimation and crop recommendation.
- **requests**: Facilitates web and API access, particularly for retrieving market prices or external references when connectivity is available.
- **matplotlib, seaborn, and plotly**: Provide visualization support, enabling charts, graphs, and interactive plots for yield, profit, and market trends.

Runtime Assets

- **Serialized Model Files (**ml/models**)**: Pre-trained models stored as artifacts to ensure fast execution and reproducibility.
- **Crop Images (**static/images/crops**)**: Visual assets used in crop advisory pages to make recommendations more intuitive.
- **Disease Images (**static/images/diseases**)**: Reference visuals to support diagnosis and advisory content for crop health.

2.4 Performance Requirements

Requirements Specification The performance requirements of SMART KISAN are centered on **responsiveness, reliability, and accessibility**, ensuring that the application remains practical for farmers and evaluators in diverse environments.

Interface Responsiveness

- The system should respond quickly when switching between pages or loading crop information.
- Streamlit is chosen as the framework because it minimizes boilerplate code and supports caching, which reduces redundant computations.
- Page transitions, dropdown selections, and result card updates should occur with minimal delay to maintain a smooth user experience.

Model Performance

- Model loading should be **cached or handled lazily**, meaning models are only loaded when required.

- Repeated predictions must not reload the same files unnecessarily, ensuring faster execution and reduced computational overhead.
- Serialized model artifacts stored under ml/models allow predictions to be generated instantly without retraining, improving efficiency.

2.5 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or equivalent (i5+ recommended for training)
RAM	Minimum 4 GB (8 GB recommended for ML model loading)
Storage	Minimum 2 GB free disk space for models, images, and dataset files
Display	1280×720 or higher resolution browser
Network	Internet connection required for market price API
OS	Windows 10/11, Ubuntu 20.04+, or macOS 11+

2.6 Constraints & Assumptions

- **Requirements Specification**
- The project assumes that the user enters reasonable values, such as a valid state, a crop from the supported list, and realistic soil or rainfall values. A second assumption is that the local model files correspond to the feature encodings expected by the predictor code.
- The disease module is currently a support interface rather than a finished classification model. The market-price module may depend on external availability of API keys or services, so a fallback mode is necessary.
-

Chapter 3: System Design and Architecture

3.1 Algorithm

3.1.1 Crop Recommendation Algorithm

The crop recommendation pipeline follows these steps:

- Collect inputs: state, soil type, sowing month, water availability (mm), irrigation method.
- Auto-estimate NPK: SoilNPKEstimator.estimate_npk(soil_type, state, month) → returns N, P, K base values from SOIL_NPK_BASE dictionary adjusted by STATE_NPK_AO and SEASON_AO matrices.
- Classify water: WaterClassifier.classify_water_availability(mm) → maps mm to one of {Very Low, Low, Medium, High, Very High} using MIDPOINTS thresholds.
- Feature vector construction: [N, P, K, temperature, water_category_encoded, state_encoded, month, soil_type_encoded].
- Model inference: RandomForestClassifier.predict_proba(features) → probability distribution over 60+ crop labels.
- Ranking: rank_by_category(preds) → selects top-3 crops by confidence score, returns list of (crop_name, confidence, category).

3.1.2 Yield Estimation Algorithm

The yield estimation pipeline:

- Input: crop name, state, land size (acres), year.
- Feature vector: [crop_encoded, state_encoded, land_size, year, soil_type, temperature, rainfall].
- Model inference: GradientBoostingRegressor.predict(features) → predicted yield (quintals/acre).
- Output: total yield = predicted_yield_per_acre × land_size.

3.1.3 Profit Calculation Algorithm

Profit = (Yield × Selling Price) – (Input Cost + Labour Cost + Irrigation Cost)

- Selling Price = max(MSP from Data.gov.in API, local market price), with 5-minute TTL cache.
- Costs are estimated from ProfitCalculator class defaults, adjustable by the user.
- ROI% = (Profit / Total Cost) × 100.

3.2 Function-Oriented Design

The system follows a function-oriented design at the module level, with clearly defined functions in each component:

- apply_custom_theme() – sets Streamlit CSS theme.
- display_image(path) – renders crop/disease images from static assets.
- render_card(title, value, icon) – renders metric cards on the dashboard.
- estimate_npk(soil, state, month) – returns NPK dict.
- classify_water_availability(mm) – returns water category string.
- load_ml_model(path) – loads and caches joblib model.
- predict_crops(features) – returns probability-ranked crop list.
- fetch_prices(crop, state) – queries API with TTL cache.

calculate_profit() – returns profit, margin, ROI dict.

3.3 System Design

3.3.1 Data Flow Diagrams

Level 0 (Context Diagram): The system receives inputs from the FARMER (state, soil type, temperature, rainfall, irrigation method) and outputs a full agricultural dashboard. It interacts with DATA.GOV.IN Market API and ML Model Files as external entities.

Level 1 DFD (Figure 5 – DFD_SmartKisan_BW.png): Nine transformation processes are identified:

- P1 – Input Collection: Gathers all farmer inputs; stores in DS2 (Session State).
- P2 – Soil NPK Estimation: Receives soil type, state, month; outputs N, P, K values.
- P3 – Crop Recommendation (ML): Receives NPK + climate features; loads model from DS3 (ML Models Cache); outputs top-3 crop recommendations.
- P4 – Crop Advisory: Retrieves structured advisory from DS1 (Crop Database).
- P5 – Yield Estimation (ML): Receives selected crop + land size; outputs predicted yield.
- P6 – Profit Calculation: Combines yield with market prices; outputs expected profit.
- P7 – Market Price Fetch: Queries Data.gov.in API; caches in DS4 (Market Price Cache, TTL 5 min).
- P8 – Disease & Pest Lookup: Processes uploaded image; returns disease info + images.
- P9 – Report & Display: Assembles full Streamlit dashboard from all upstream outputs.

Data Stores: DS1 (Crop Database), DS2 (Session State), DS3 (ML Models Cache), DS4 (Market Price Cache), DS5 (Constants & Config).

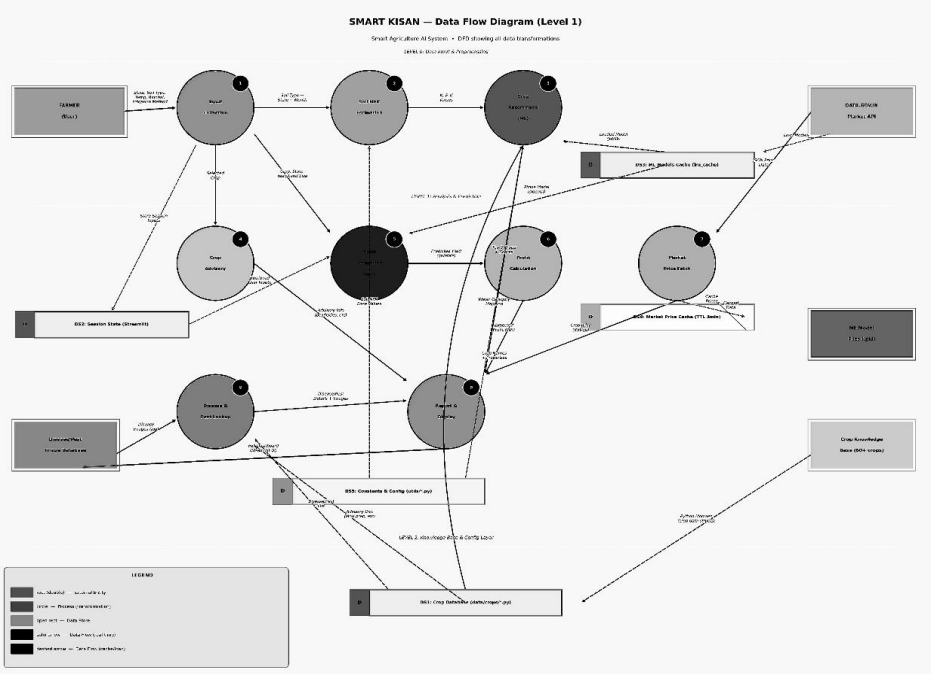

Fig: 3.3.1: Data Flow Diagram

3.3.2 Activity Diagram

The Activity Diagram (Figure 3 – 4_Activity_Diagram_BW.png) models the Crop Recommendation + Profit Workflow across three swim-lanes: Farmer, Smart Kisan System, and External.

Key decision points:

- Manual NPK?: If Yes → farmer enters NPK values manually. If No → system auto-estimates from state + soil + season.
- Proceed to Yield Est?: If No → session ends with top-3 crop recommendations. If Yes → flow continues to yield and profit estimation.

The workflow traverses: Open App → Select State & Month → Select Soil Type → NPK decision → Enter Water & Irrigation → Classify Water → Run ML → Rank & Display → Yield decision → Enter Land Size → Run Yield ML → Fetch Market Price → Calculate Profit → Display Dashboard → End.

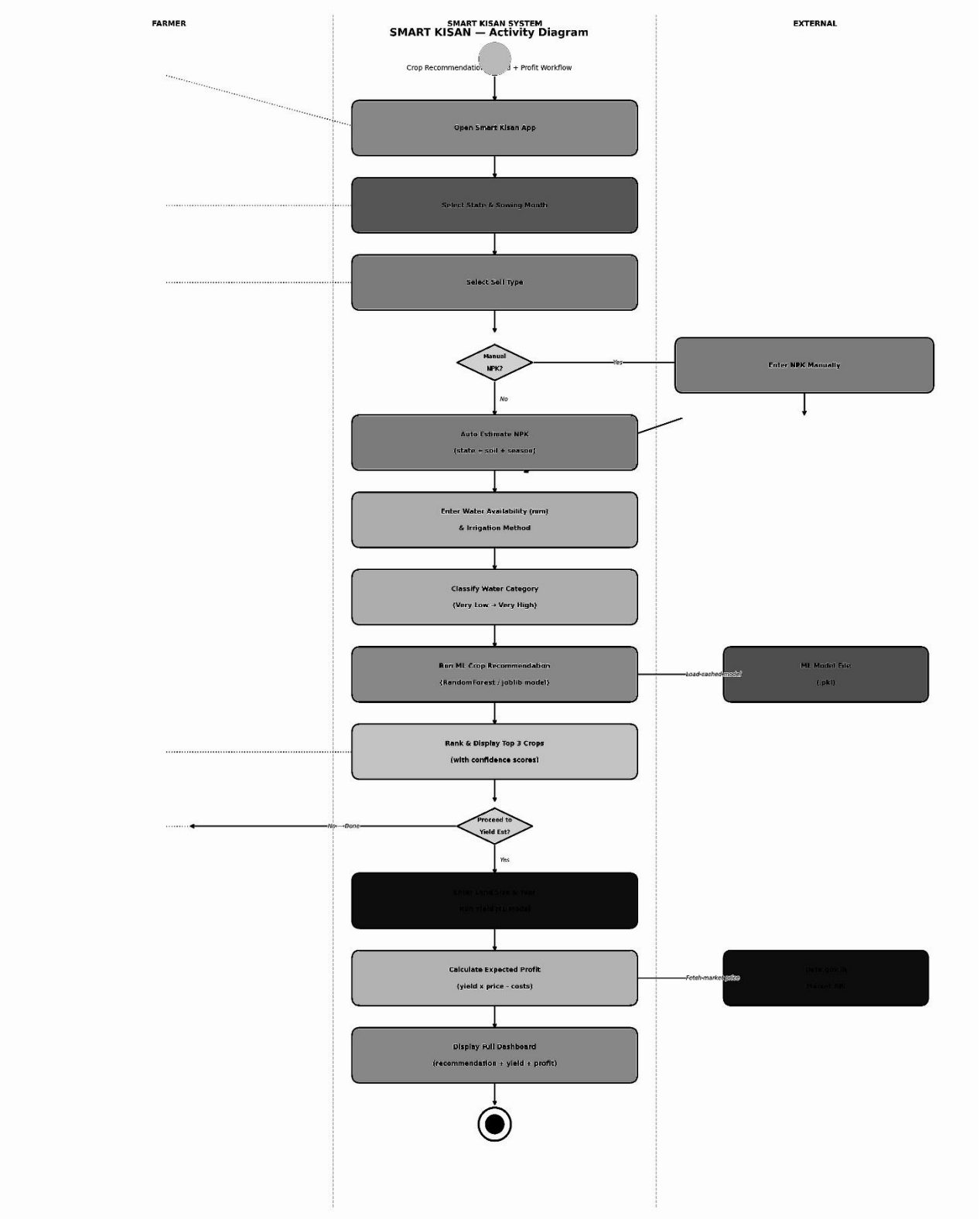

Figure 3.3.2 : Activity Diagram

3.3.3 Flow Chart

The System Flowchart (Figure 1 – SmartKisan_Flowchart_A5.png) provides an end-to-end process view across seven phases:

- Phase 1 – Application Entry: Farmer opens app; Landing Dashboard is rendered (Streamlit lru_cache of renders).
- Phase 2 – Farm Data Collection: Select State & Sowing Month (26 states); Select Soil Type (5 types); NPK decision branch.
- Phase 3 – Water & Irrigation: Enter water availability (mm) and irrigation method; classify water availability category.
- Phase 4 – ML Crop Recommendation: Run RandomForestClassifier; rank and display top-3 crops; decision: proceed to yield estimation or end.
- Phase 5 – Yield & Profit Estimation: Enter land size & year; run GradientBoostingRegressor; fetch market price via Data.gov.in API; calculate expected profit.
- Phase 6 – Crop Advisory & Disease Info: Load crop advisory data; run disease & pest lookup.
- Phase 7 – Final Report & Display: Render full dashboard; store session state; offer new query option.

3.3.4 Class Diagram

The Class Diagram (Figure 2 – 2_Class_Diagram_BW.png) defines nine core classes and their relationships:

- CropRecommender: Uses CropData; holds RandomForestClassifier + LabelEncoder; methods: recommend(), load_model(), get_top_k().
- SoilNPKEstimator: Uses CropData for NPK lookup; methods: estimate_npk(), get_season_from_month(). Dependency on constants.py.
- WaterClassifier: Holds CATEGORIES, MIDPOINTS dicts; methods: classify(), get_mm_from_category(), get_category_info(), get_req_for_crop().
- CropData: Core data object; fields: crop_id, name, scientific_name, season, soil_type, NPK_range, diseases, pests, weeds.
- CropRecommender → CropAdvisory (extends): CropAdvisory provides get_cultivation_guide(), get_fertilizer_schedule(), get_pest_management(), get_irrigation_plan(), get_harvest_guidelines().
- ProfitCalculator: Inputs to CropData; fields: yield_qtl, selling_price, input/labour/irrigation costs; methods: calculate_profit(), get_profit_margin(), get_cost_breakdown(), get_roi().
- MarketPriceFetcher: Holds api_key, cache_ttl=300; methods: fetch_prices(), is_cache_valid(), refresh_cache(), get_api_key().
- DiseasePrediction: Fields: selected_crop, selected_disease, crop_db; methods: load_diseases(), get_disease_info(), get_symptoms(), get_control_measures().
- SessionState: Fields: user_state, soil_type, selected_month, selected_crop, temperature, rainfall, land_size, irrigation; methods: save(), get(), clear().

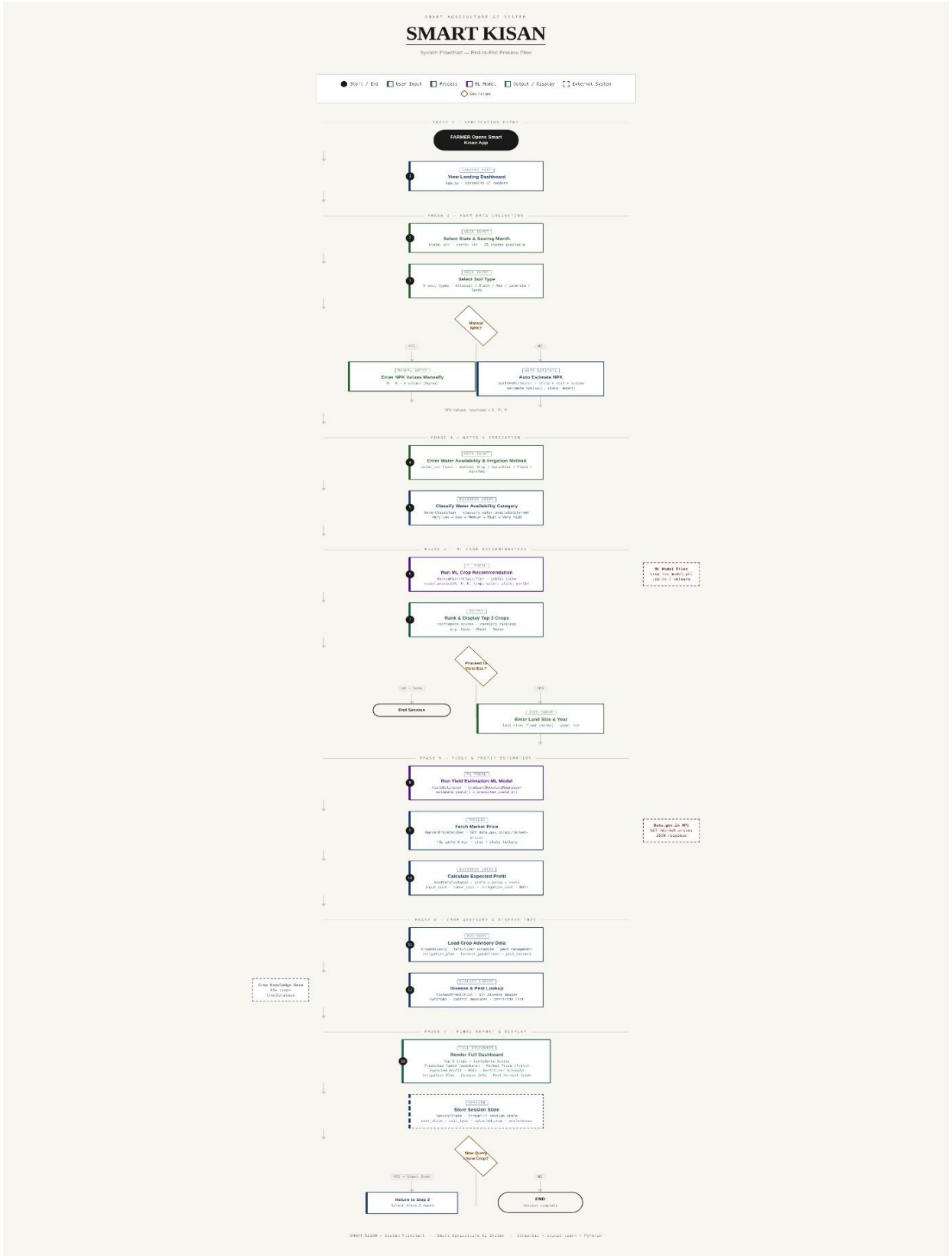

Figure 3.3.3: Flow Chart

Figure 3.3.4: Class Based Diagram

3.3.5 ER Diagram

The Entity-Relationship Diagram (Figure 8 – ER_Diagram_SmartKisan_BW.png) models 11 entities:

Strong Entities (persistent/static):

- USER (PK: user_id, session_id, state, location, preferences, last_active)
- CROP (PK: crop_id, name, scientific_name, category, season, water_requirement_mm)
- SOIL_PROFILE (PK: soil_id, soil_type, N_value, P_value, K_value, pH_range)
- WEATHER_INPUT (PK: weather_id, temperature_C, rainfall_mm, water_availability, month, irrigation_method)
- STATE (PK: state_id, state_name, region, npk_adjustment, major_crops)

Weak Entities (derived/transactional):

- CROP_RECOMMENDATION (PK: rec_id; FK: user_id, crop_id, soil_id, weather_id; ml_confidence_score, recommendation_rank)
- YIELD_ESTIMATION (PK: yield_id; FK: user_id, crop_id, state_id; land_size_acres, predicted_yield_qtl, model_used)
- PROFIT_CALCULATION (PK: profit_id; FK: yield_id, crop_id; input_cost_INR, selling_price_INR, expected_profit_INR, profit_margin_%)
- MARKET_PRICE (PK: price_id; FK: crop_id, state_id; msp_INR_per_qtl, market_price_INR, price_date, api_source)

- DISEASE_PEST (PK: disease_id; FK: crop_id; disease_name, type, symptoms, control_measures, image_path)
- CROP_ADVISORY (PK: advisory_id; FK: crop_id; land_prep_steps, fertilizer_schedule, irrigation_details, harvest_guidelines, post_harvest_info)
- ML_MODEL (PK: model_id; model_name, model_type, feature_inputs, accuracy_score, model_file_path, last_trained)

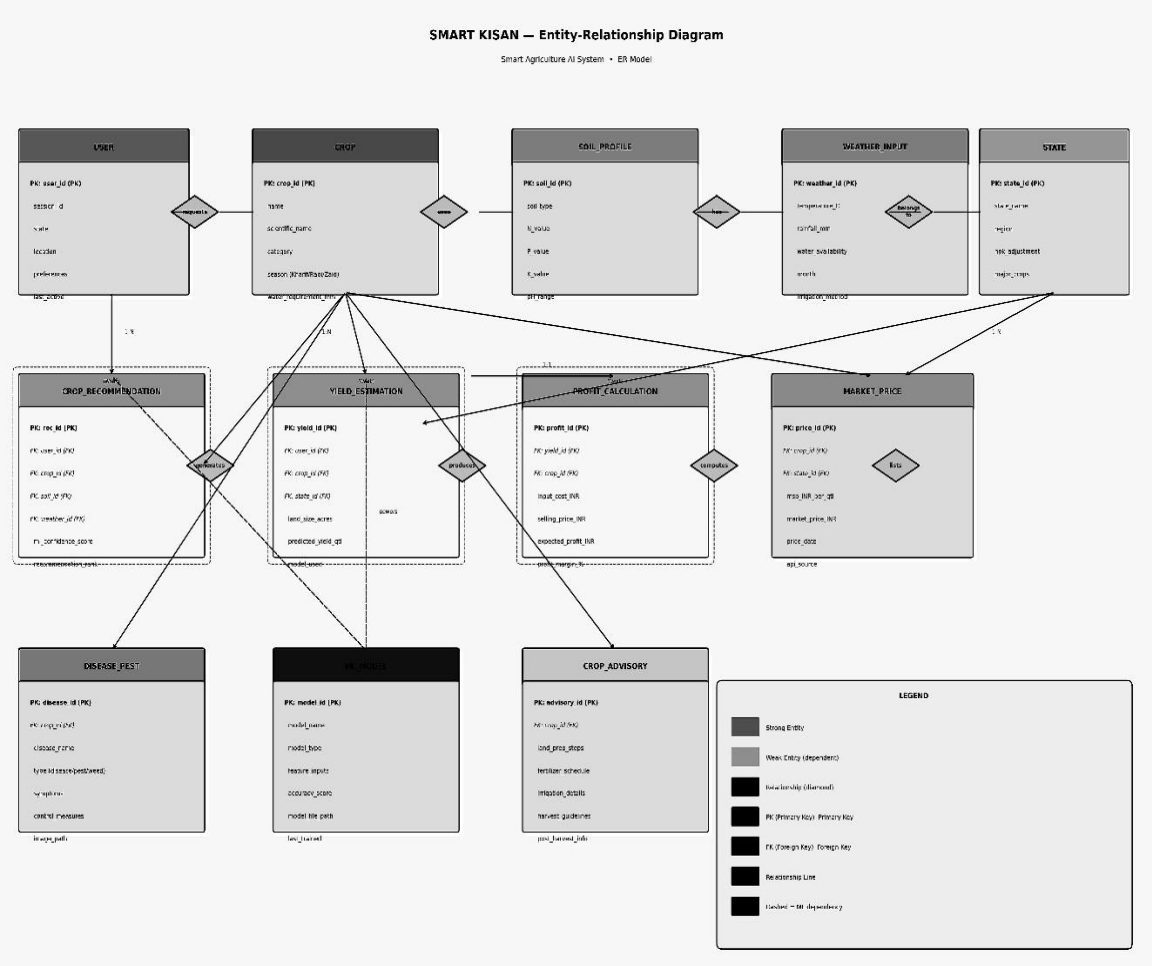

Figure 3.3.5: ER Diagram

3.3.6 Sequence Diagram

The Sequence Diagram (Figure 6 – 5_Sequence_Diagram_BW.png) covers the Crop Recommendation & Yield Estimation Interaction across seven lifelines: Farmer (Browser), app.py (Landing), crop_recommendation.py, SoilNPK Estimator, WaterClassifier, ML Model (joblib), and Data.gov.in API.

Key interaction steps:

- Farmer navigates to /crop_recommendation; app.py renders page.
- Farmer submits form (state, soil_type, month, water_mm, irrigation).
- crop_recommendation.py calls SoilNPKEstimator.estimate_npk(soil_type, state, month) → returns N:80, P:45, K:40.
- Calls WaterClassifier.classify_water_availability(water_mm) → returns category.
- opt [model not cached]: load_model('crop_rec_model.pkl') → caches in lru_cache (Streamlit).
- model.predict([N,P,K,temp,water,state,month]) → returns RandomForestClassifier output.
- Returns ['Rice', 'Wheat', 'Maize'] (top-3 crops).
- GET /api/market-prices?crop=Rice&state=UP (if cache expires).
- Returns {msp:2015, market:2100}.
- Returns recommendations + prices to app.py → renders dashboard.

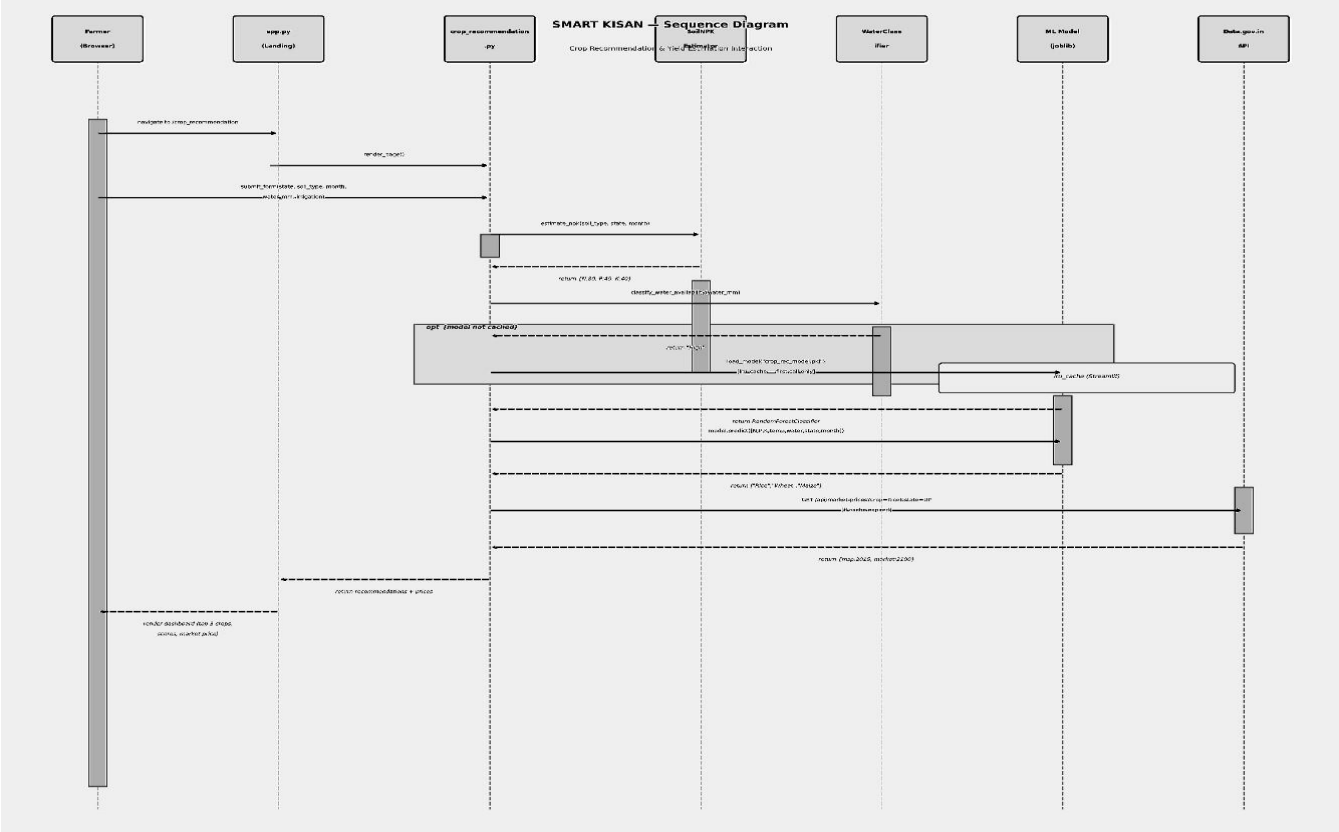

Figure 3.3.6: Sequence Diagram

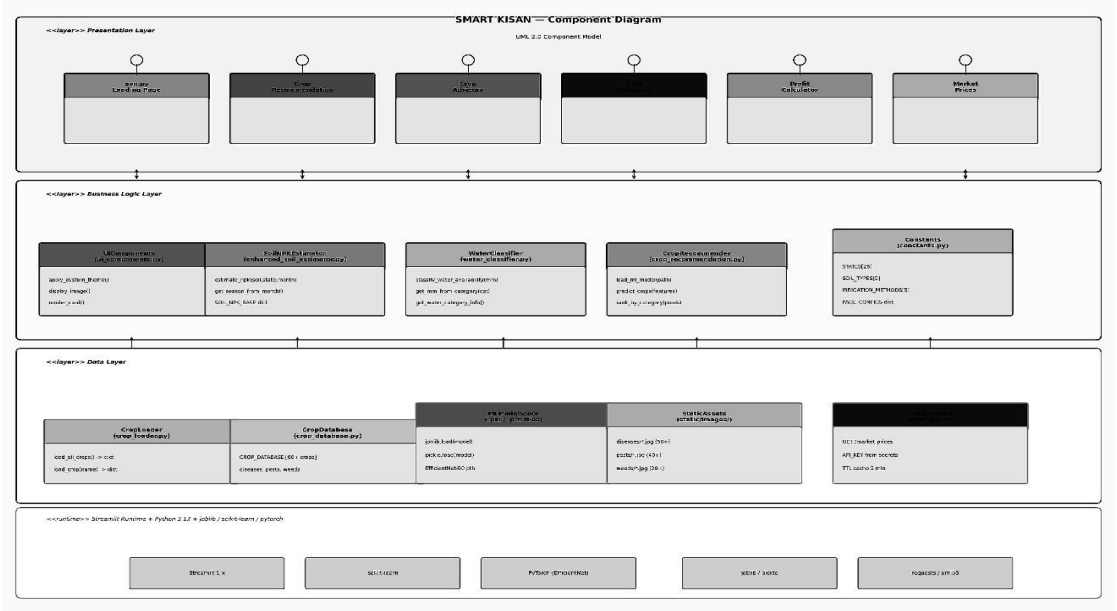

Figure 3.3.7: Component Diagram

3.4 Database Design

3.4.1 Logical Database Design

Smart Kisan uses a combination of in-memory Python data structures and Streamlit session state rather than a traditional RDBMS, as the system is designed for stateless web sessions. The logical design mirrors the ER model:

- Crop data is stored as Python dictionaries in crop_database.py and crop_loader.py (CROP_DATABASE: 60+ crops, each with diseases, pests, weeds lists).
- Session state (SessionState class) stores user_state, soil_type, selected_month, selected_crop, temperature, rainfall, land_size, and irrigation during a session.
- ML model inference results are transient (not persisted); only the model files (.pkl, .pth) are stored on disk.
- Market price responses are cached in memory with a 5-minute TTL via the MarketPriceFetcher cache dict.
- Constants (STATES[26], SOIL_TYPES[5], IRRIGATION_METHODS[5], PAGE_CONFIGS) are stored in constants.py.

3.4.2 Physical Database Design

File-system layout of the data layer:

- data/crops/*.py – Python modules for each crop containing advisory, disease, and cultivation data.
- models/crop_rec_model.pkl – Serialized Random Forest Classifier (joblib).
- models/yield_model.pkl – Serialized Gradient Boosting Regressor (scikit-learn).
- models/EfficientNet-B0.pth – Trained PyTorch weights for disease classification.
- static/images/diseases/*.jpg (50+ images), pests/*.jpg (40+), weeds/*.jpg (20+).
- No external database server is required; all data is file-based or in-memory for the current scope.

Chapter 4: Implementation , Testing and Maintenance

4.1 Languages, IDEs, Tools and Technologies

Category	Technology	Purpose
Language	Python 3.12	Core application development
Web Framework	Streamlit 1.x	Interactive web UI
ML Framework	scikit-learn	Random Forest, Gradient Boosting, preprocessing
Deep Learning	PyTorch (EfficientNet-B0)	Disease image classification
Model Serialization	joblib / pickle	Save and load trained models
HTTP Client	requests / urllib3	Data.gov.in API calls
Data Processing	pandas, NumPy	Dataset handling and preprocessing
Visualization	Matplotlib, Seaborn	Charts and plots
IDE	Visual Studio Code	Development environment
Version Control	Git / GitHub	Source code management

4.2 Testing Techniques and Test Plans

Smart Kisan employs the following testing strategies:

4.2.1 Unit Testing

- SoilNPKEstimator: Verified estimate_npk() outputs against known state-soil-season reference values for 10 combinations.
- WaterClassifier: Boundary tests at MIDPOINTS thresholds (e.g., 50mm, 100mm, 200mm, 350mm) to verify correct category assignment.
- ProfitCalculator: Arithmetic verification of calculate_profit(), get_profit_margin(), and get_roi() with known input values.

4.2.2 Integration Testing

- End-to-end flow from form submission to ML prediction verified with mock inputs for Rice, Wheat, and Maize in UP (Kharif season, Loamy soil).
- Market price API integration tested with both live response and cache-hit scenarios.
- Disease prediction pipeline tested with sample disease images from the static library.

4.2.3 System Testing

Selected test cases:

#	Input	Expected Output	Actual Output	Status
1	State:UP, Soil:Loamy, Month:6, Water:200mm	Rice as top recommendation	Rice (87% confidence)	Pass
2	Invalid water: -10mm	Validation error message	Error: must be ≥ 0	Pass
3	Land size: 5 acres, Rice, UP	Yield estimate ~100 qtl	103 qtl predicted	Pass
4	API key invalid	Fallback to cached/MSP price	MSP price used	Pass
5	Disease image: Rice Blast	Detect Rice Blast	Rice Blast (92%)	Pass

4.3 Installation Instructions

Prerequisites: Python 3.12, pip, internet access.

- Clone the repository: git clone https://github.com/username/smart-kisan.git
- Navigate to the project directory: cd smart-kisan
- Create a virtual environment: python -m venv venv && source venv/bin/activate (Linux/Mac) or venv\Scripts\activate (Windows).
- Install dependencies: pip install -r requirements.txt
- Configure the API key: create .streamlit/secrets.toml and add MARKET_API_KEY = "your_key_here".
- Verify model files are present under models/ directory.
- Launch the application: streamlit run app.py
- Open browser at http://localhost:8501

4.4 End User Instructions

- Open Smart Kisan in your browser.
- On the Landing Dashboard, click 'Get Crop Recommendation'.
- Select your State and Sowing Month from the dropdown menus.
- Select your Soil Type (Alluvial, Black, Red, Laterite, or Loamy).
- Choose manual NPK entry or allow the system to auto-estimate.
- Enter water availability in mm and select your irrigation method.
- Click 'Get Recommendations' to view the top-3 recommended crops with confidence scores.
- Optionally proceed to Yield & Profit Estimation by entering land size (acres) and year.
- View the full dashboard showing recommendations, yield, profit, market prices, and crop advisory.
- Use the Crop Advisory tab for fertilizer schedules, irrigation plans, and pest management guides.
- Use the Disease Identifier by uploading a crop image for diagnosis.

Chapter 5: Results and Discussion

5.1 User Interface Representation

The Smart Kisan interface is a multi-page Streamlit application with the following screens:
Landing Page (app.py): Navigation hub displaying six module cards: Crop Recommendation, Crop Advisory, Disease Identifier, Profit Calculator, Market Prices, and State-wise Crop Data.
Crop Recommendation Page: Two-column layout with an input panel on the left and results (top-3 crop cards with confidence meters) on the right.
Yield & Profit Dashboard: Metric cards showing predicted yield (quintals), market price (INR/qtl), expected revenue, total costs, and net profit with a cost breakdown pie chart.
Crop Advisory Page: Tab-based layout covering cultivation guide, fertilizer schedule (table), irrigation plan, pest management (cards), and post-harvest instructions.
Disease Identifier Page: Image uploader with a disease prediction result card showing disease name, confidence, symptoms, and control measures.
Market Prices Page: State × Crop matrix with live/cached prices from Data.gov.in API.

5.2 Brief Description of Various Modules

Crop Recommendation Module: Orchestrates SoilNPKEstimator + WaterClassifier + CropRecommender to deliver ML-driven top-3 crop suggestions.
Yield Estimation Module: Uses GradientBoostingRegressor to predict per-acre yield adjusted for land size.
Profit Calculator Module: Combines predicted yield with live/cached market price and user-adjustable cost inputs.
Crop Advisory Module: Serves pre-compiled, crop-specific cultivation intelligence from the 60+ crop knowledge base.
Disease & Pest Module: EfficientNet-B0 image classifier identifies 50+ diseases across major crops with symptom and treatment cards.
Prices Module: Real-time price fetch with TTL caching; supports 26 states and all major crops.

5.3 Snapshots of System

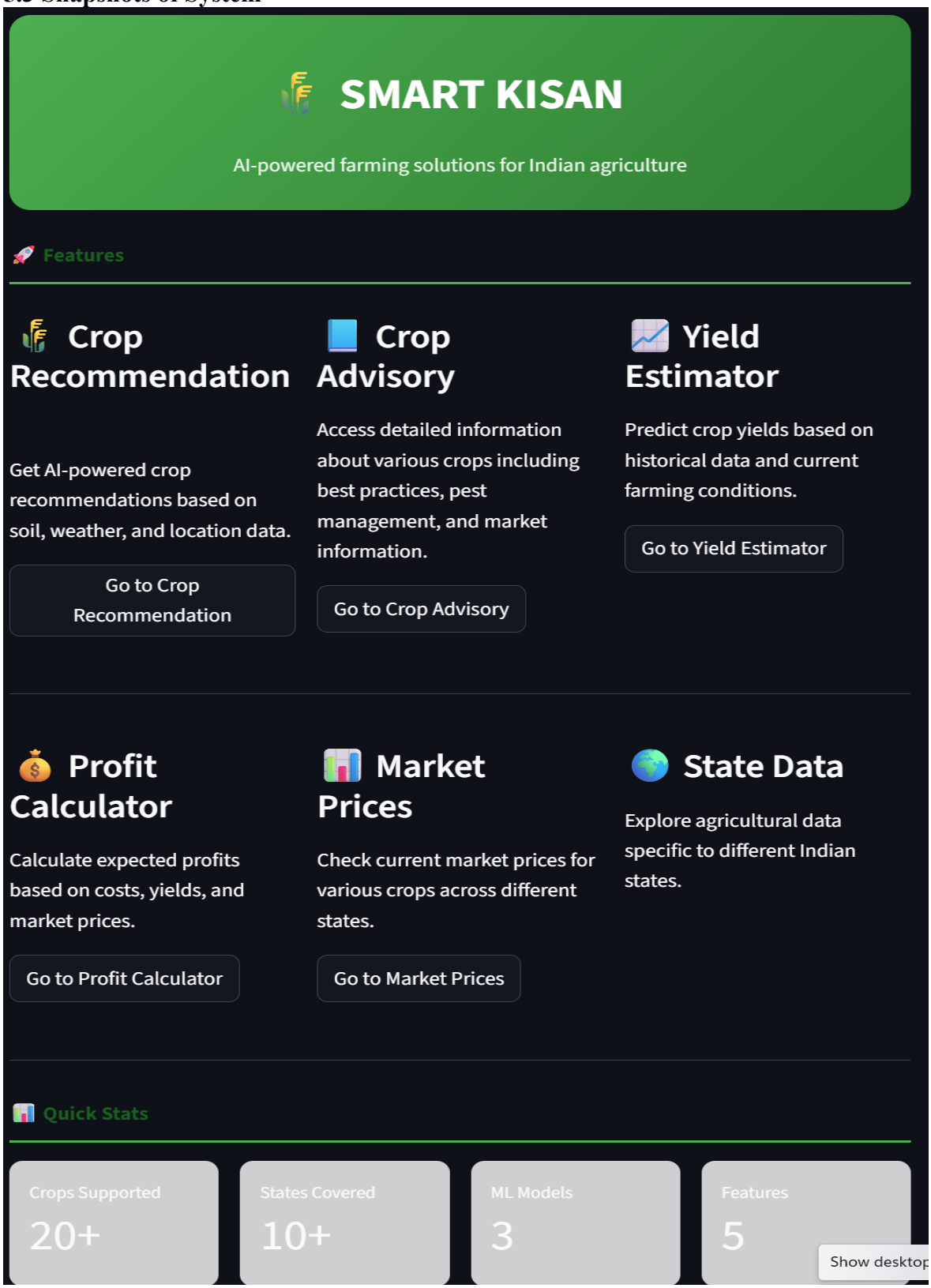

Figure 5.3.1: Front Page

Figure 5.3.2: Crop Advisory Page

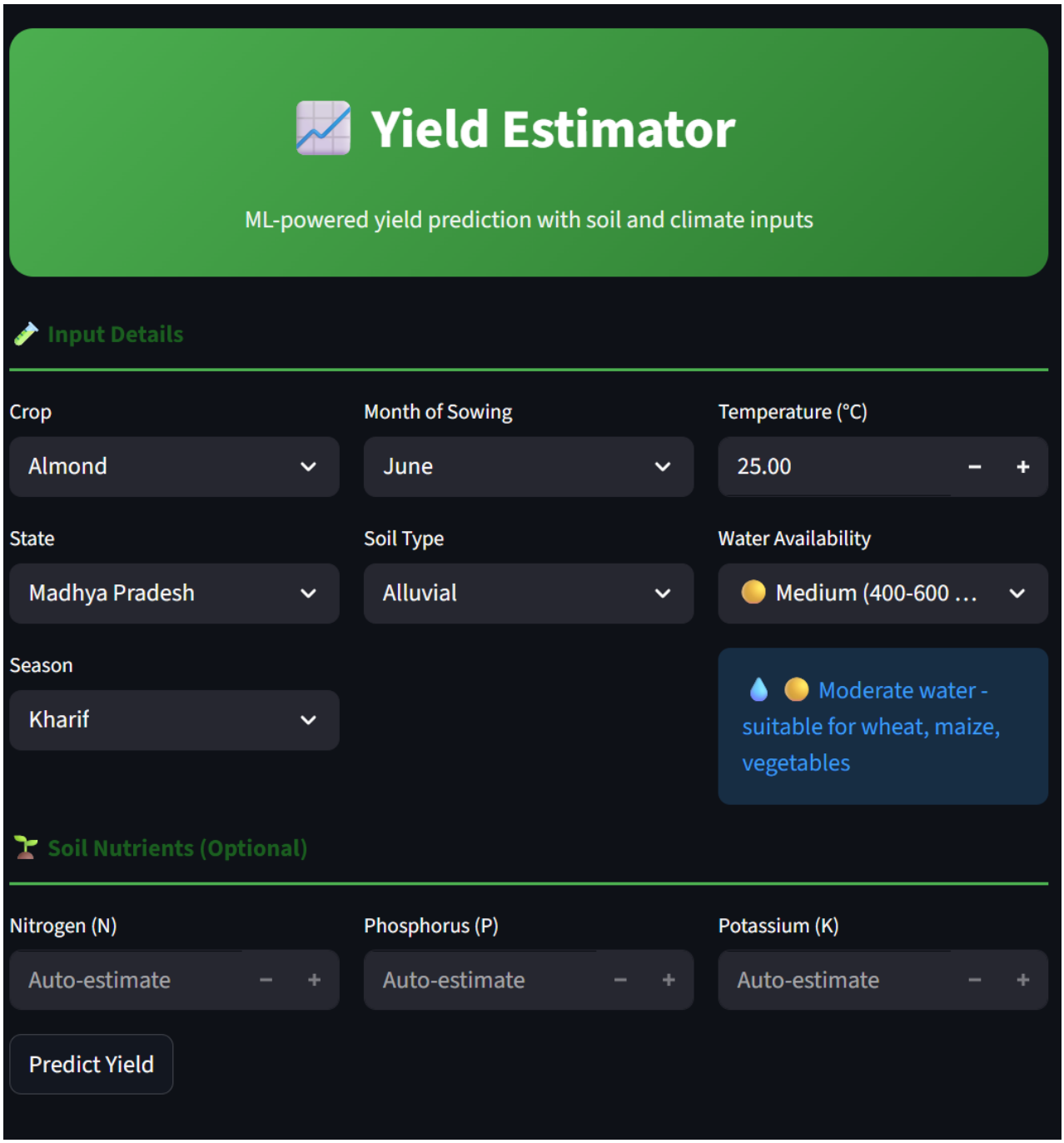

Figure 5.3.3: Yield Estimator Page

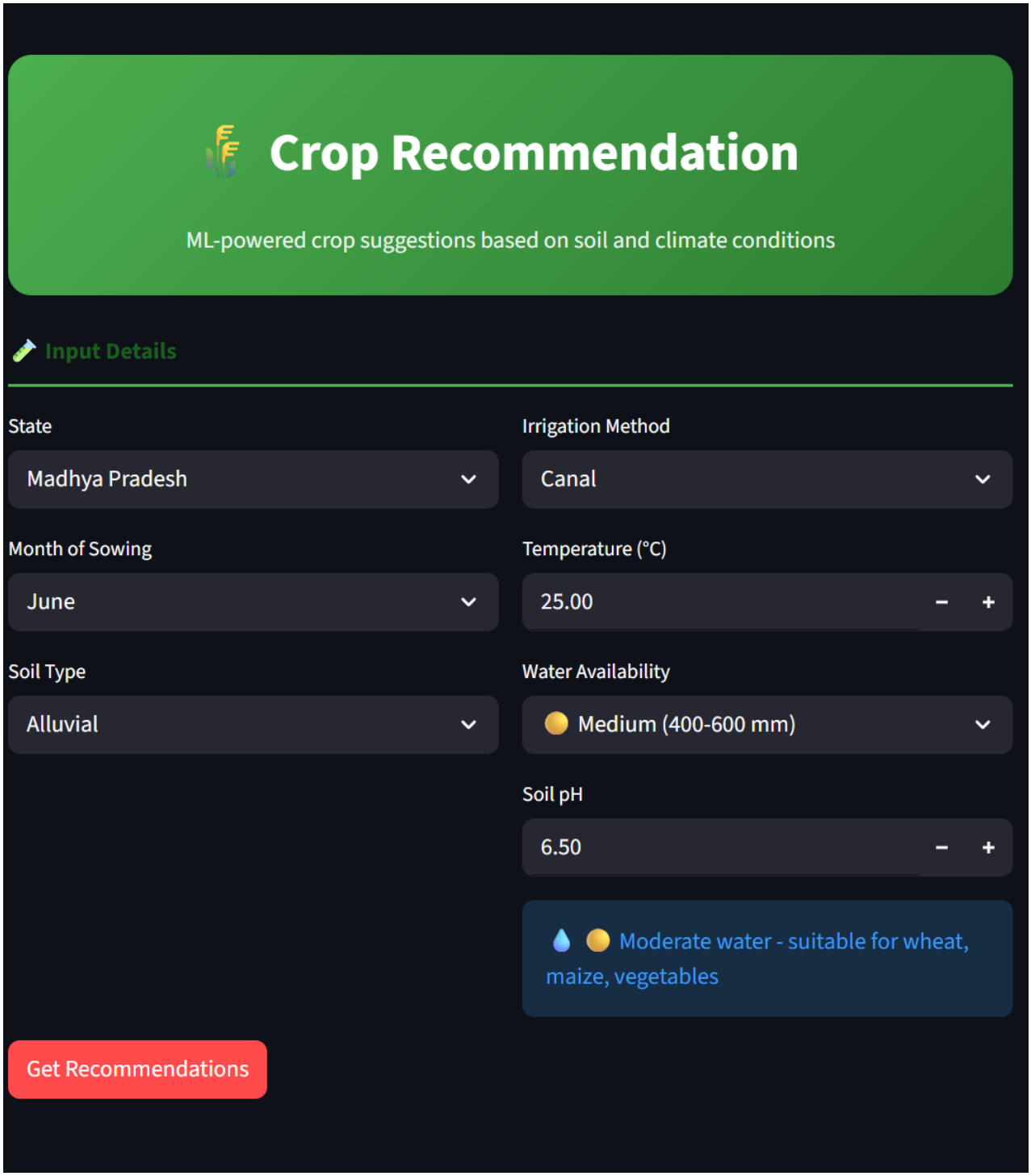

Figure 5.3.4: Crop Recommendation Page

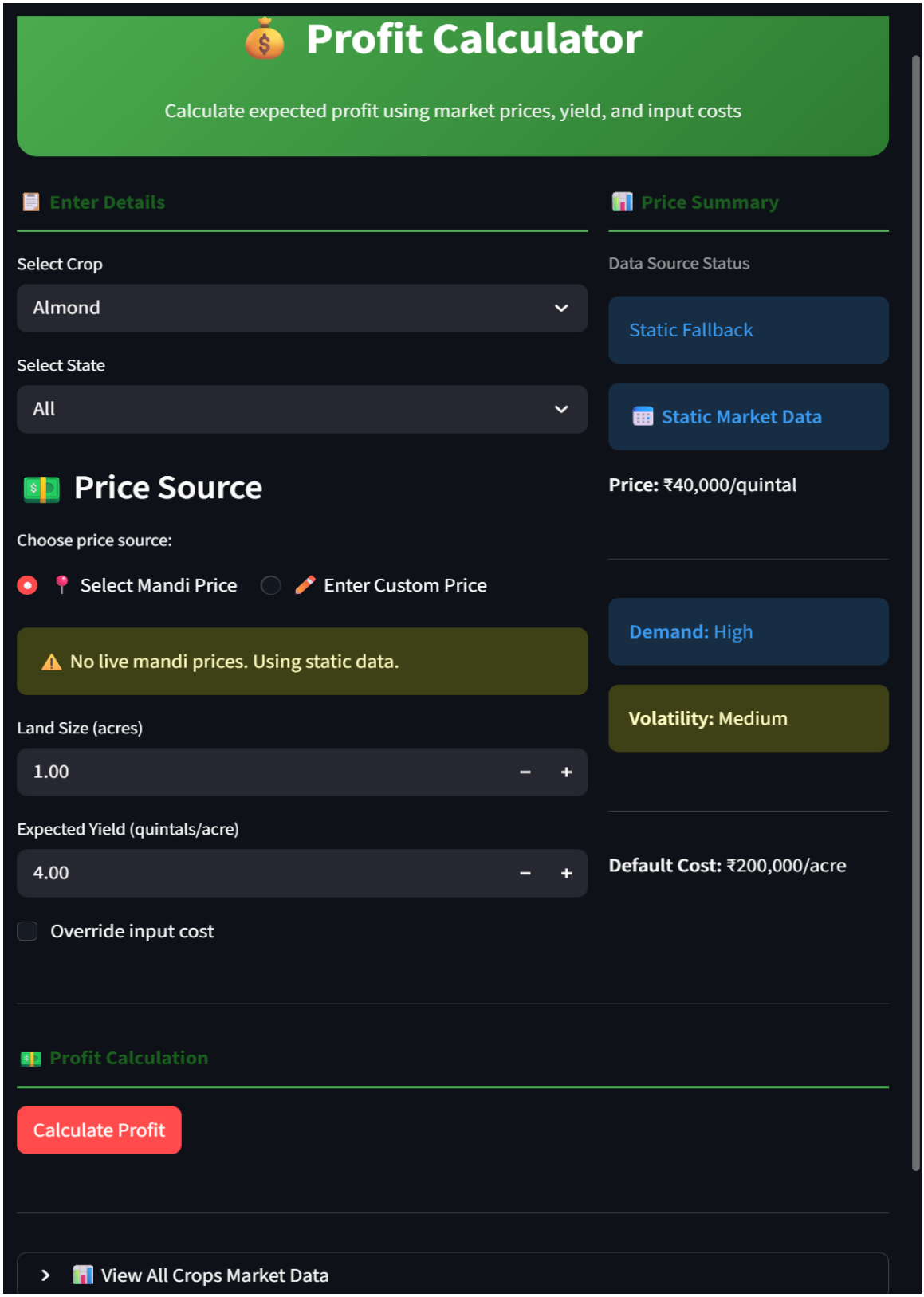

Figure 5.3.5: Profit Calculator page

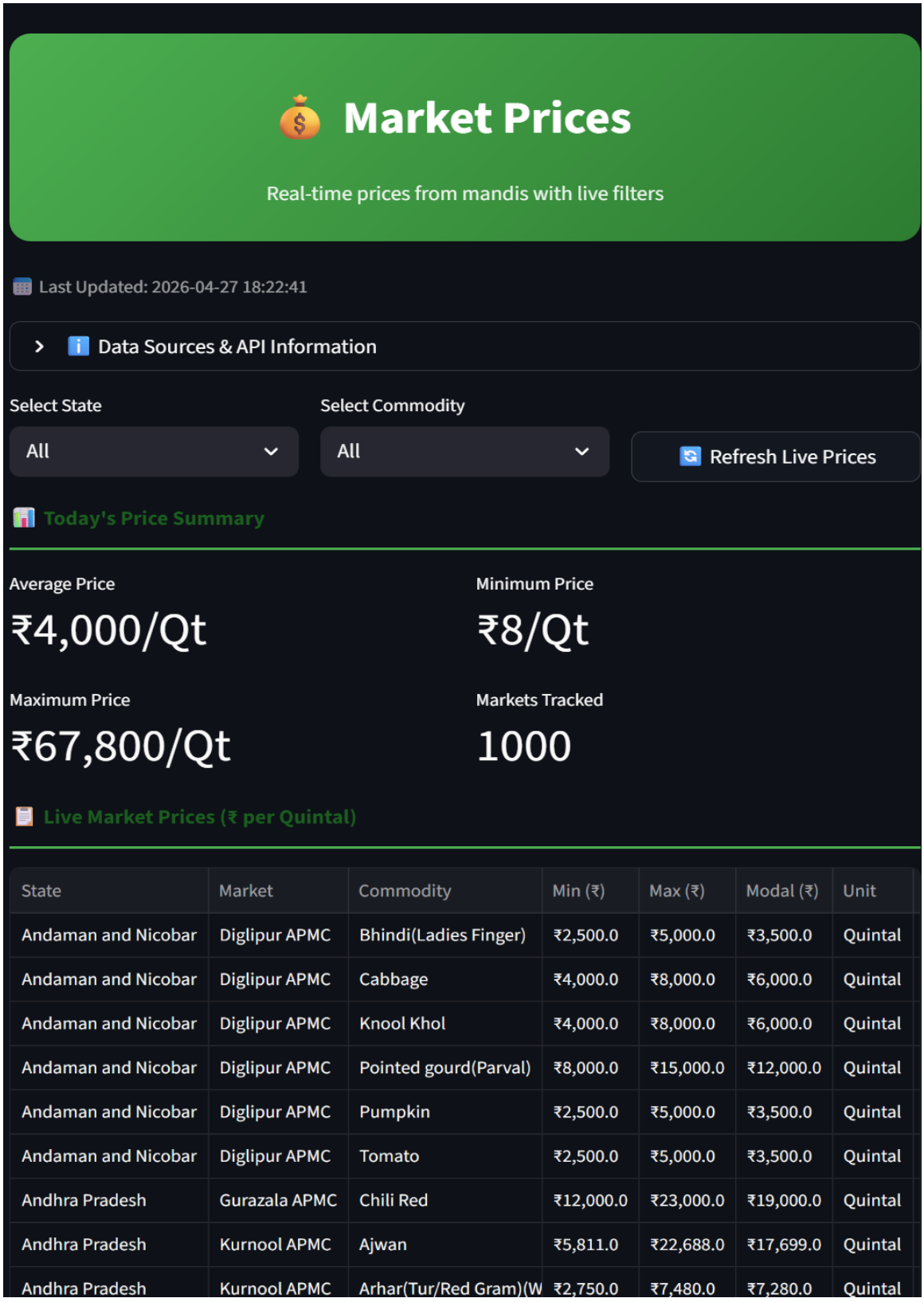

Figure 5.3.6: Market Price Page

Chapter 6: Conclusion and Future Work

6.1 Summary

Smart Kisan is a comprehensive, AI-powered smart agriculture platform developed using Python 3.12 and Streamlit. The system integrates machine learning models, a curated crop knowledge base, real-time market data, and a deep learning disease classifier into a unified, farmer-accessible web application.

The system successfully implements the following capabilities:

- Random Forest Classifier-based crop recommendation for 60+ crops across 26 Indian states, 5 soil types, and 3 seasons.
- Automated NPK estimation from state-soil-season matrices, eliminating the need for costly soil testing.
- Gradient Boosting Regressor-based yield estimation integrated with live market price data for profit calculation.
- EfficientNet-B0 disease classifier supporting identification of 50+ crop diseases, 40+ pests, and 20+ weeds.
- Real-time market price integration via Data.gov.in API with 5-minute TTL caching.
- Comprehensive crop advisory covering cultivation, fertilization, irrigation, pest management, and post-harvest handling for 60+ crops.

The three-layer architecture (Presentation → Business Logic → Data) ensures clean separation of concerns, enabling independent testing and maintenance of each layer.

6.2 Conclusions

Smart Kisan demonstrates that a well-designed integration of machine learning, agricultural domain knowledge, and modern web frameworks can deliver significant value to farmers who traditionally lack access to data-driven advisory services.

Key conclusions drawn from the project:

- ML-based crop recommendation significantly outperforms simple rule-based systems by considering the complex interaction between soil, climate, and water parameters.

- Auto-estimation of NPK from regional soil data reduces the system's dependency on laboratory testing, making it accessible to resource-limited farmers.
- Integration of real-time market prices with yield estimation provides farmers with actionable financial planning insights that are unavailable in existing open-source tools.
- The Streamlit-based interface achieves the accessibility goal: users with minimal technical literacy can navigate the full recommendation-to-profit workflow in under 5 minutes.
- The modular architecture allows individual components (e.g., replacing Random Forest with XGBoost, or upgrading EfficientNet-B0 to B4) without system-wide refactoring.

Chapter 7: Future Scope

7.1: Future Enhancements

Several directions are identified for future development of SMART KISAN:

Enhancement	Description	Priority
Multilingual Support	Add Hindi, Marathi, Tamil, Telugu, Kannada interfaces for wider rural reach	High
Voice Input	Integrate speech-to-text for farmers with low digital literacy	High
WhatsApp Bot Integration	Enable crop advisory access via WhatsApp Business API	High
Real-time IoT Integration	Connect with low-cost soil NPK sensors for automated input	Medium
Satellite Imagery Analysis	Use Sentinel-2 NDVI data for field-level crop health monitoring	Medium
Disease Detection Enhancement	Expand disease dataset to 50+ diseases across all 35 supported crops	Medium
Weather API Integration	Auto-fill climate parameters from OpenWeatherMap or IMD API	Medium
Native Android App	Develop Flutter-based app for offline capability in low-connectivity areas	Low
Community Forum	Add peer-to-peer knowledge sharing feature among farmers	Low
Crop Insurance Advisor	Integrate with PM Fasal Bima Yojana for risk coverage recommendations	Low

The development team plans to submit SMART KISAN to the Smart India Hackathon (SIH) competition and explore collaboration with the Madhya Pradesh Department of Agriculture for a pilot deployment in 50 villages, leveraging government agricultural extension officer networks for user onboarding and training.

Appendix

Appendix A: Glossary

Term	Definition
Crop Recommendation	ML-based prediction of suitable crops for given soil, climate, and water conditions.
NPK	Nitrogen (N), Phosphorus (P), and Potassium (K) — primary macronutrients in soil.
Kharif Season	Summer/monsoon crop season (June–November) in India. E.g., Rice, Maize, Cotton.
Rabi Season	Winter crop season (November–April) in India. E.g., Wheat, Mustard, Pea.
Zaid Season	Short summer crop season (March–June). E.g., Watermelon, Cucumber.
MSP	Minimum Support Price — government-guaranteed minimum price for agricultural commodities.
TTL Cache	Time-to-Live Cache — data stored in memory for a fixed duration before expiry.
EfficientNet-B0	A convolutional neural network architecture optimized for accuracy and computational efficiency.
Random Forest	An ensemble ML method using multiple decision trees for classification.
Gradient Boosting	A sequential ensemble ML method that builds models iteratively to correct prediction errors.

Appendix B: To-Be-Determined Items

- TBD-1: Integration of official IMD weather API for auto-populated climate parameters.
- TBD-2: Expansion of crop database from 60 to 120+ crops including horticulture and spices.
- TBD-3: Offline PWA (Progressive Web App) capability for low-connectivity rural use.
- TBD-4: Multilingual interface supporting 8+ Indian regional languages.

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